

Lecture 12:

Isochrones and age structure

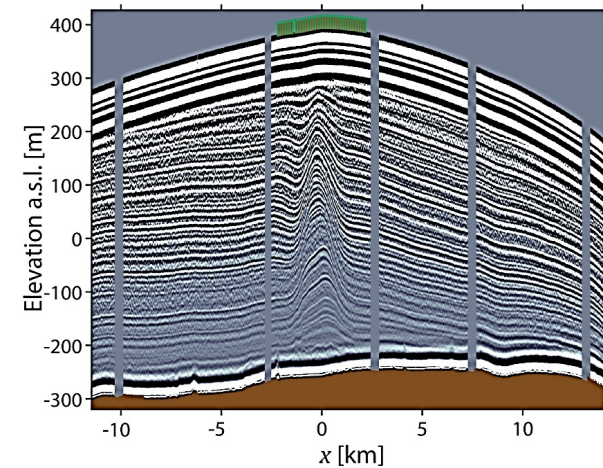
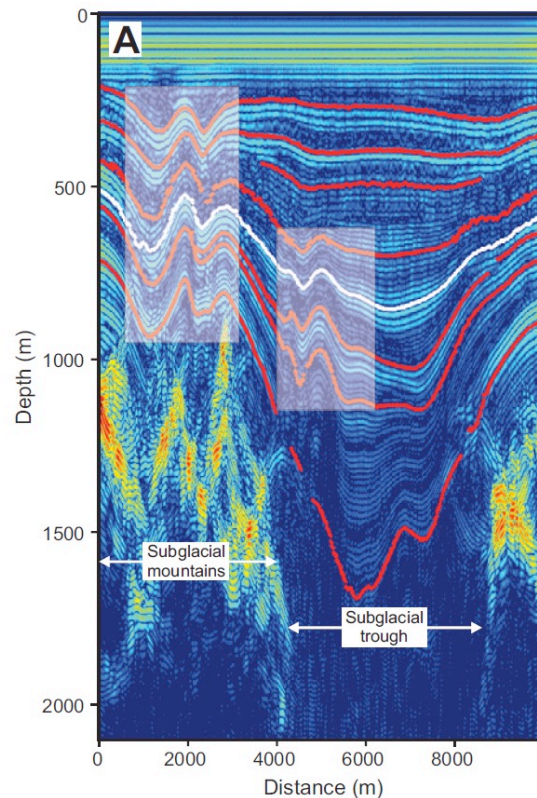
Radar layers

Isochrones

The age equation

Theoretical isochrones

Applications



Ice-penetrating radar



Photo Credit: Iain Rudkin

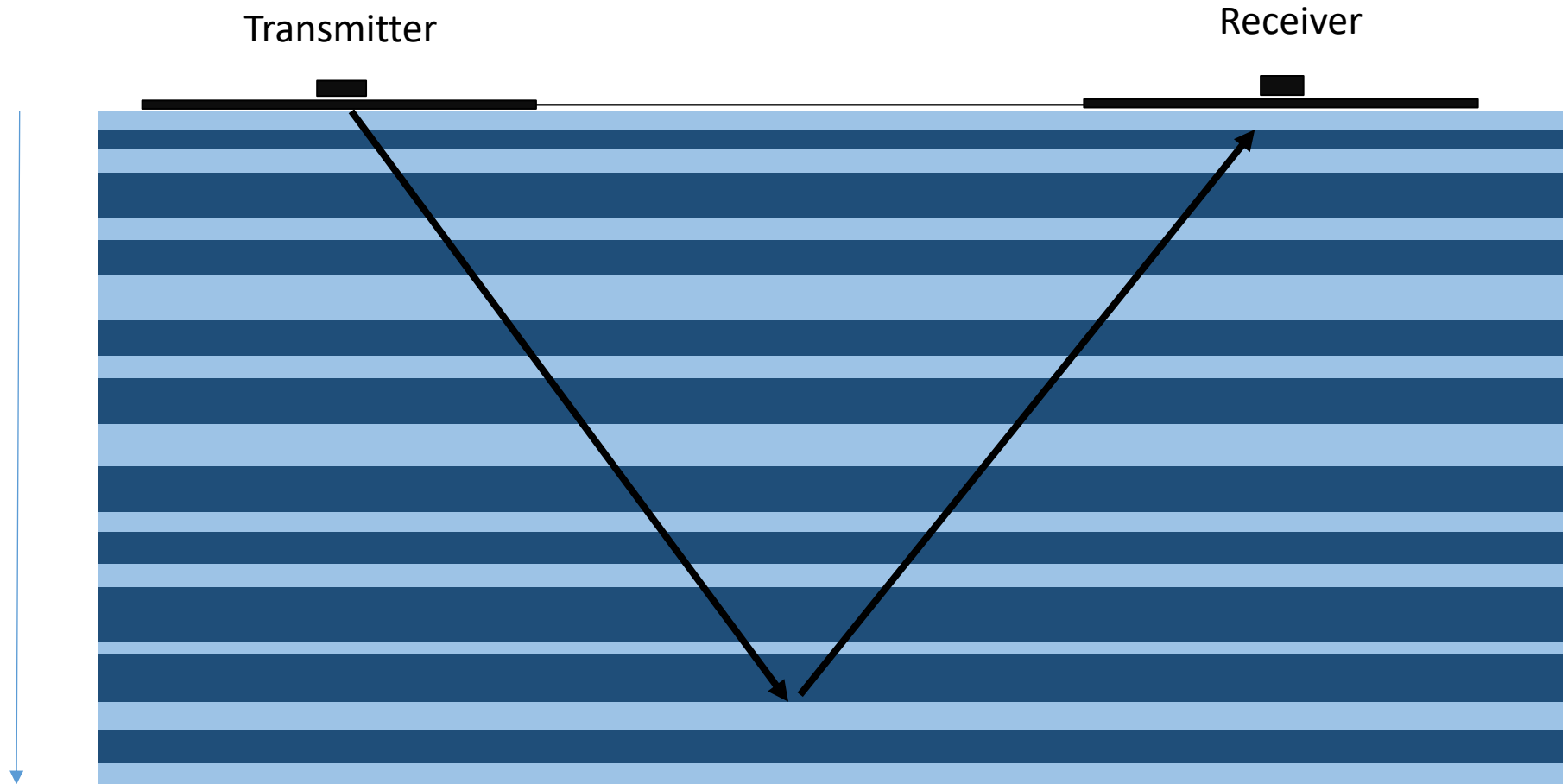
Ice-penetrating radar



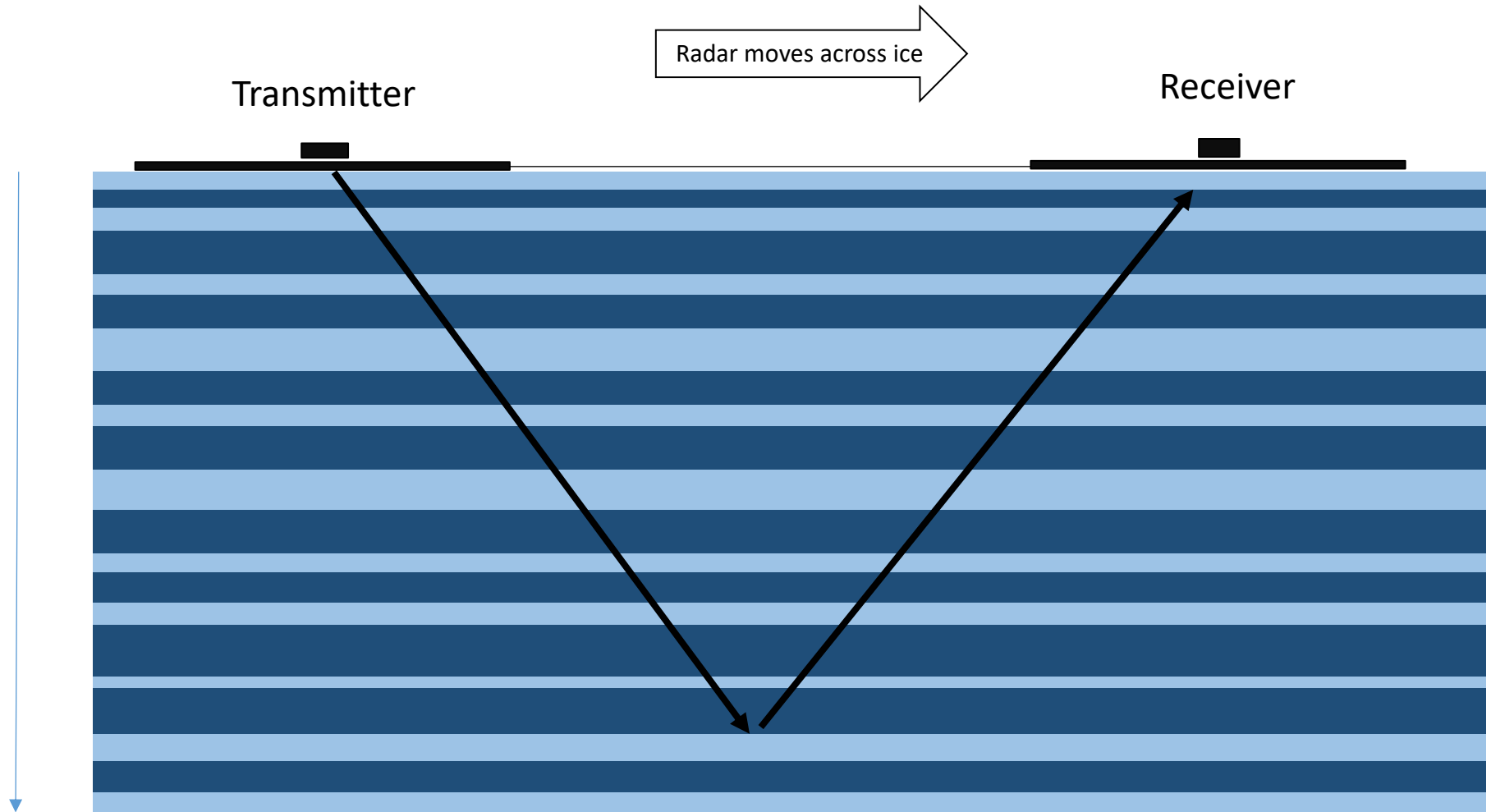
Ice-penetrating radar



Ice-penetrating radar

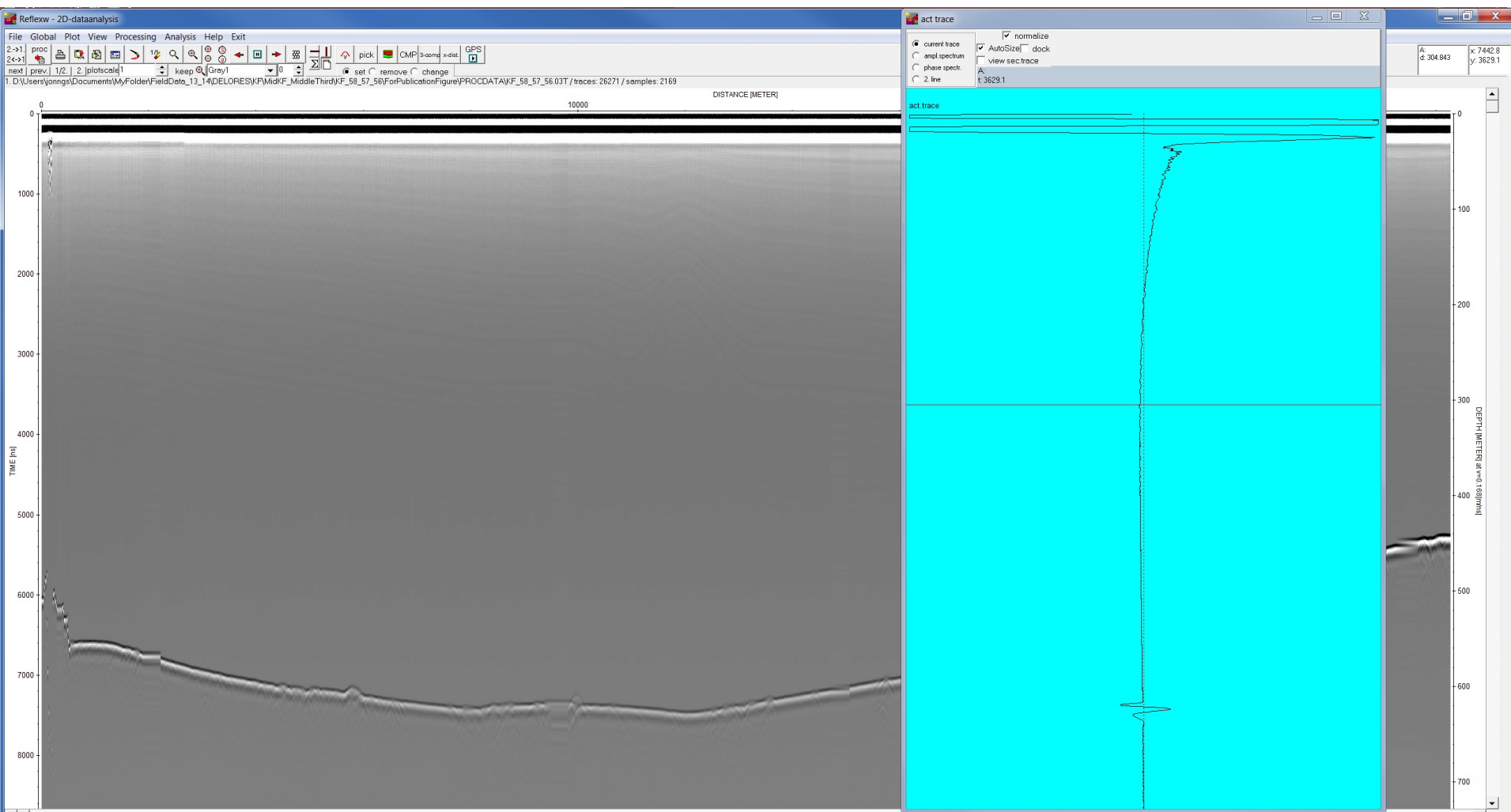


Ice-penetrating radar



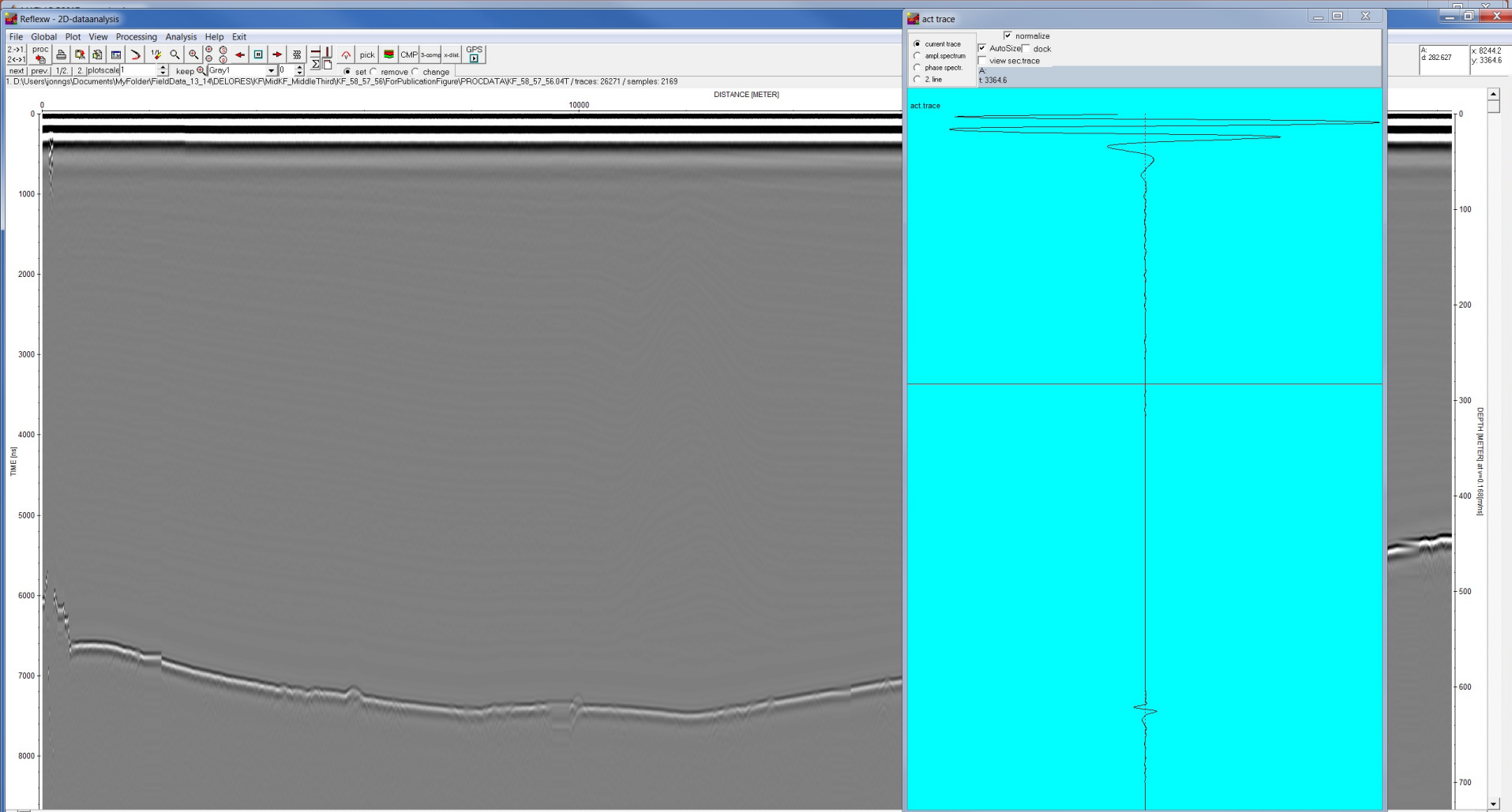
Ice-penetrating radar

Raw data from ground-based ice-penetrating radar



Ice-penetrating radar

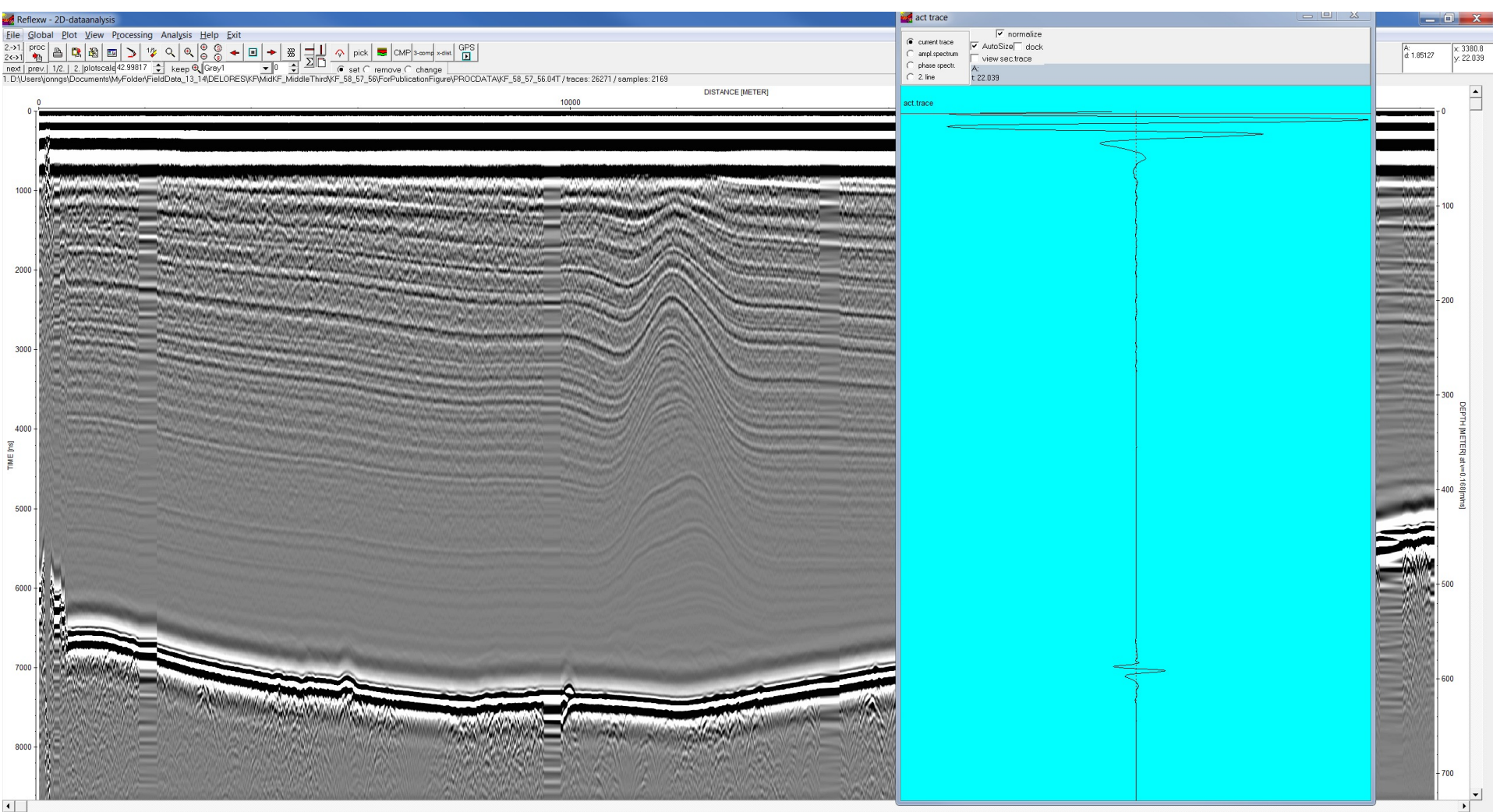
Filtered data from ground-based ice-penetrating radar



<https://www.youtube.com/watch?v=spUNpyF58BY>

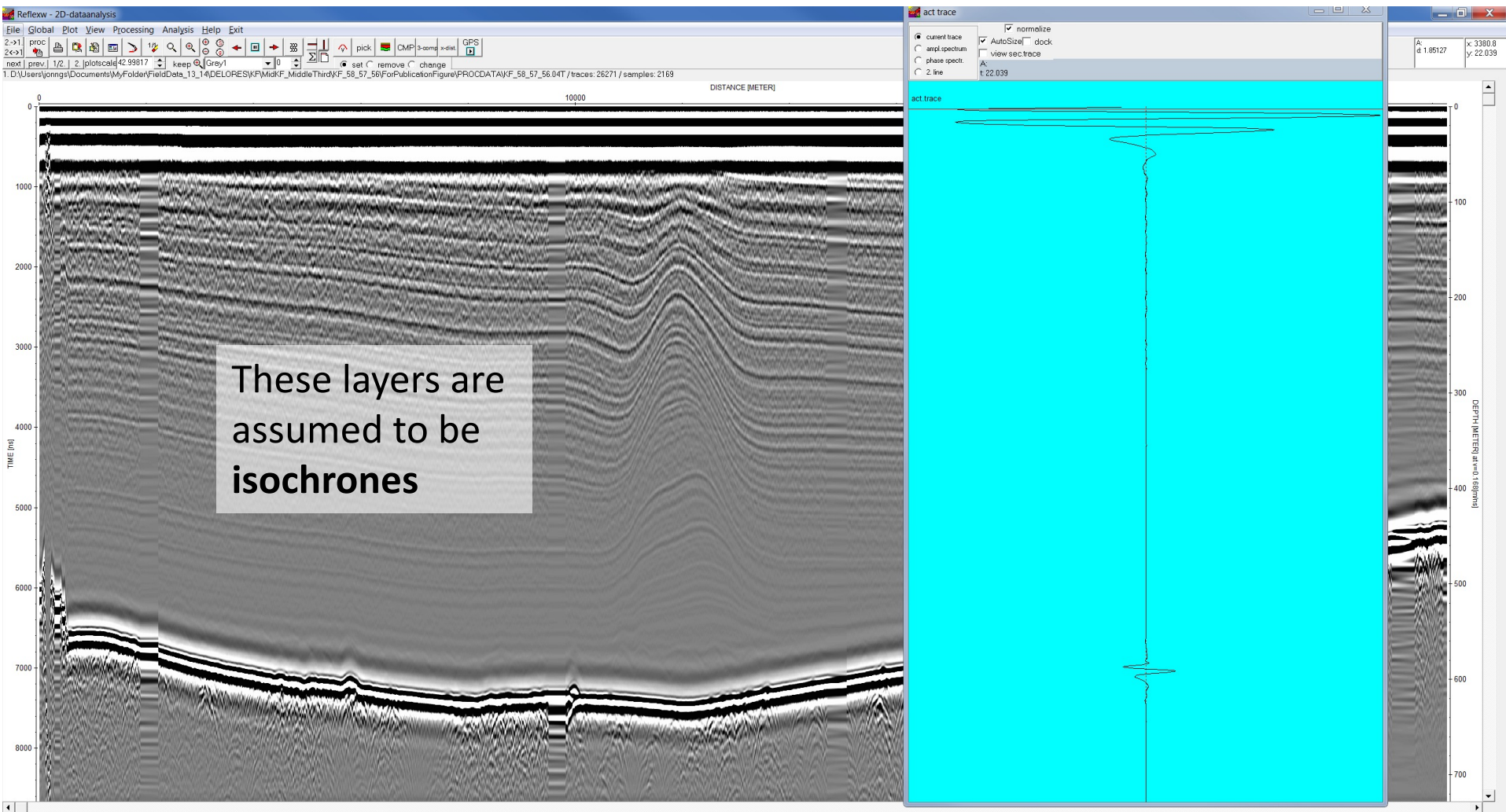
Ice-penetrating radar

Filtered data from ground-based ice-penetrating radar, contrast stretched.

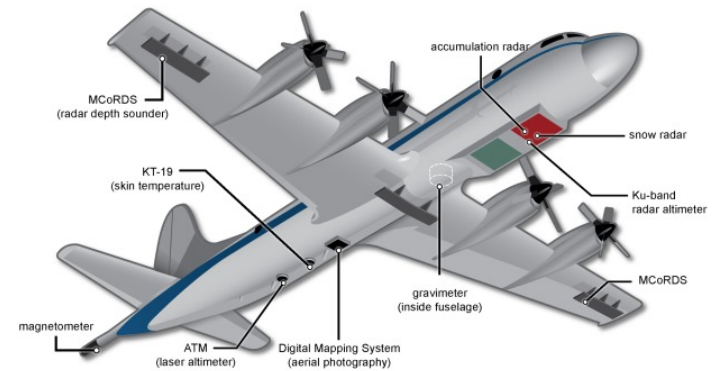


Ice-penetrating radar

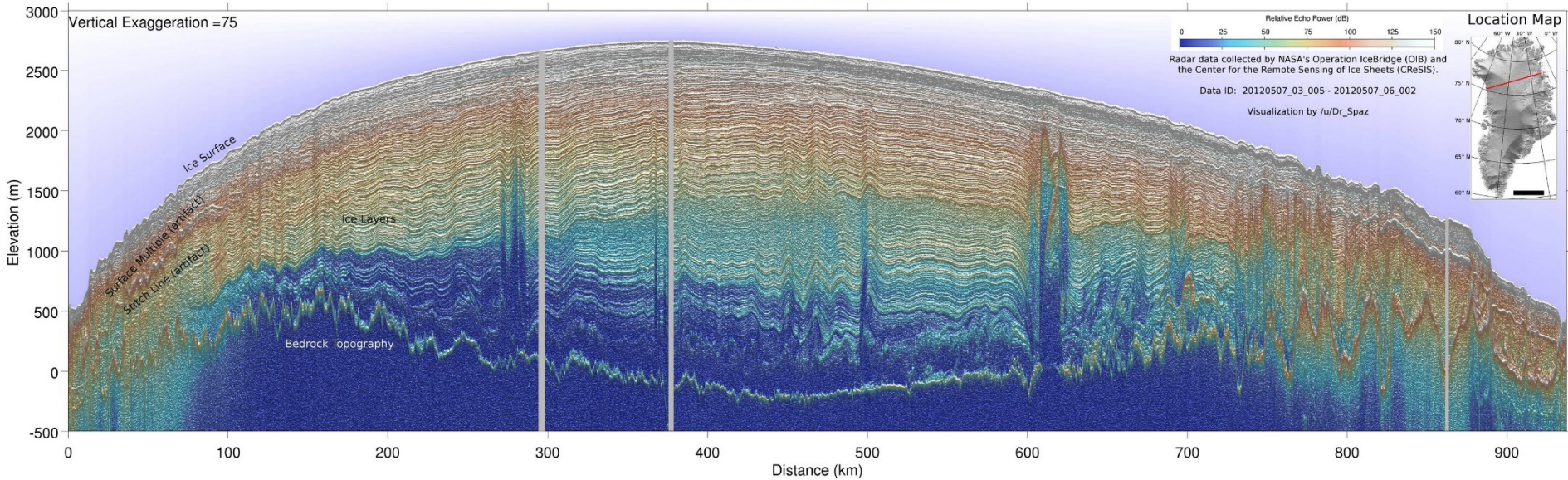
Filtered data from ground-based ice-penetrating radar

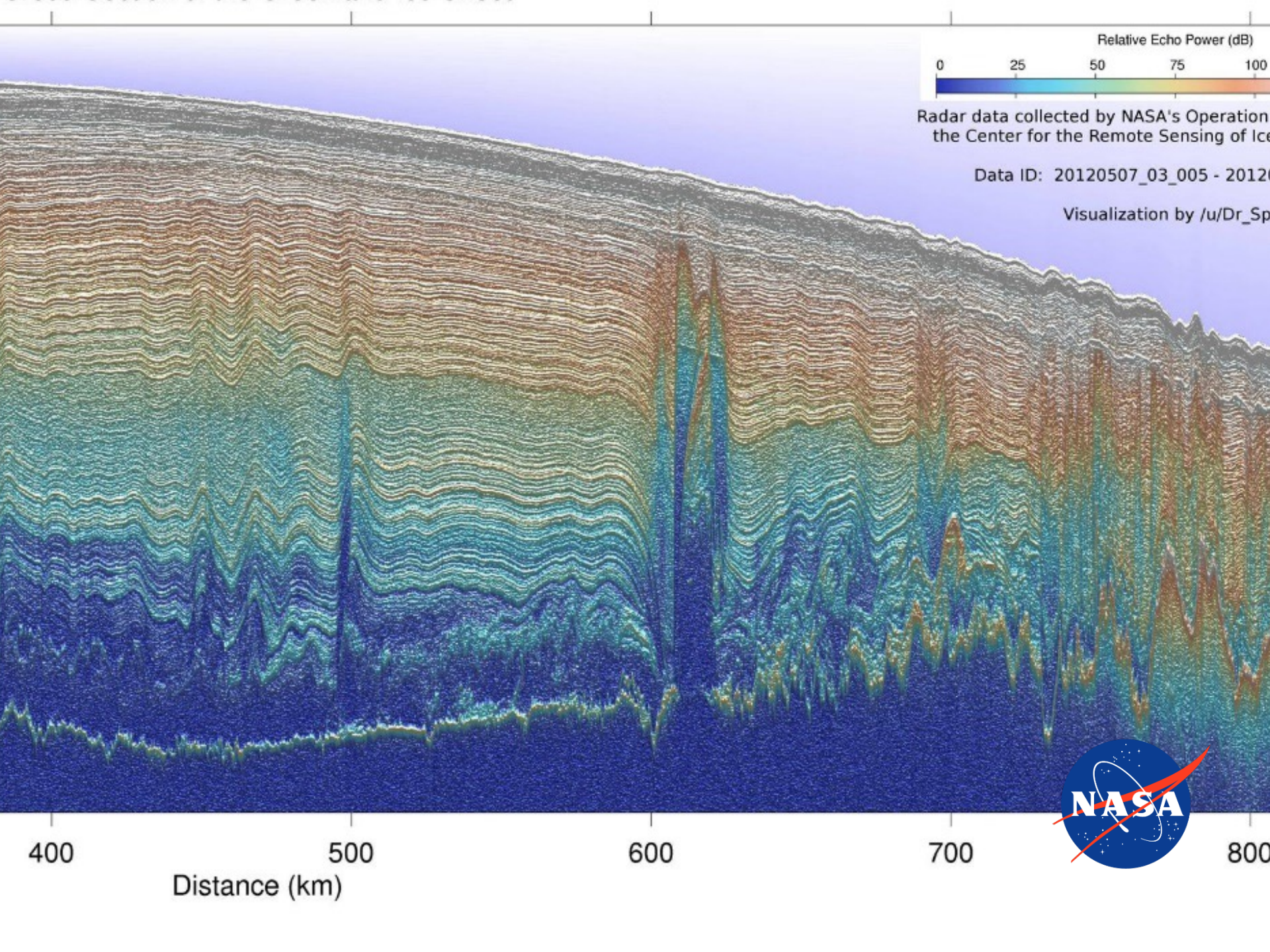


Ice-penetrating radar



Cross-Section of the Greenland Ice Sheet

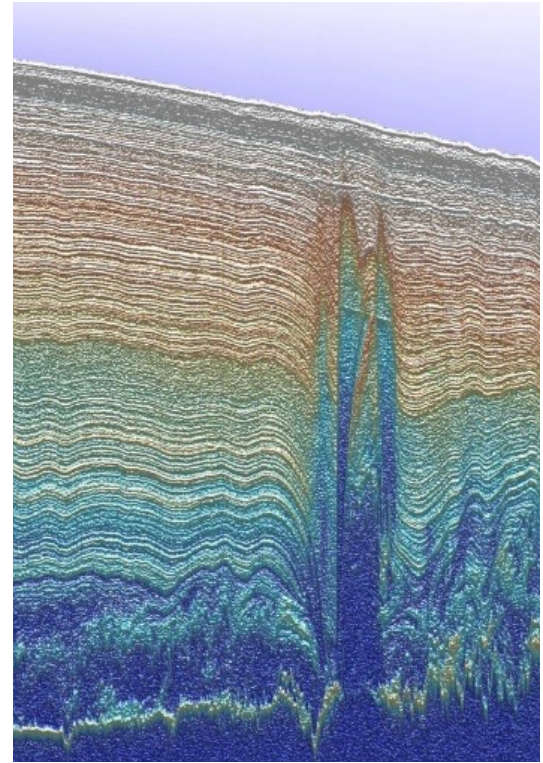




Greenland's age structure from isochrones

The Age Equation

How does age evolve?



The Age Equation

How does age evolve?

We treat age as a scalar field

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Lagrangian coordinate
system

(moves with the ice as it flows)

Eulerian coordinate system

(stays stationary in space)

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$$\frac{DA}{Dt} = 1$$

The Age Equation

How does age evolve?

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Lagrangian coordinate system

(moves with the ice as it flows)

$$\frac{DA}{Dt} = 1$$

Eulerian coordinate system

(stays stationary in space)

$$\frac{\partial A}{\partial t} + \mathbf{u} \cdot \nabla A = 1$$

The Age Equation

How does age evolve?

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In 1D

$$\frac{\partial A}{\partial t} + w \frac{\partial A}{\partial z} = 1$$

The Age Equation

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$$\frac{\partial A}{\partial t} + w \frac{\partial A}{\partial z} = 1$$

In 1D, in
steady state

$$\frac{\partial A}{\partial z} = \frac{1}{w}$$

$$A(z = H) = 0$$

Solve this in the same way as we solved the heat equation

```
def age(p = 3, a_mperyear = 0.1):
    # time domain
    dt_years = 5 # time step in years
    seconds_per_year = 365*24*60*60 # approximate number of seconds in year
    dt = dt_years*seconds_per_year # time step in seconds
    time_total_years = 6e4 # total length of the simulation in years
    T = time_total_years*seconds_per_year; # total length of the simulation in seconds
    time = np.linspace(0,T,round(T/dt)) # the time grid, units [years]

    # space domain
    grid_spacing = 10 # grid spacing, units [m]
    ice_thickness = 753; # The max depth of the Roosevelt Age depth profile
    depth = np.linspace(0,ice_thickness,round(ice_thickness/grid_spacing)) # spatial grid, units [m]

    # vertical velocity
    accumulation_rate = a_mperyear/seconds_per_year
    vertical_velocity = -accumulation_rate*(1-(p+2)/(p+1)*(depth/ice_thickness) + 1/(1+p) *(depth/ice_thickness)**(p+2))

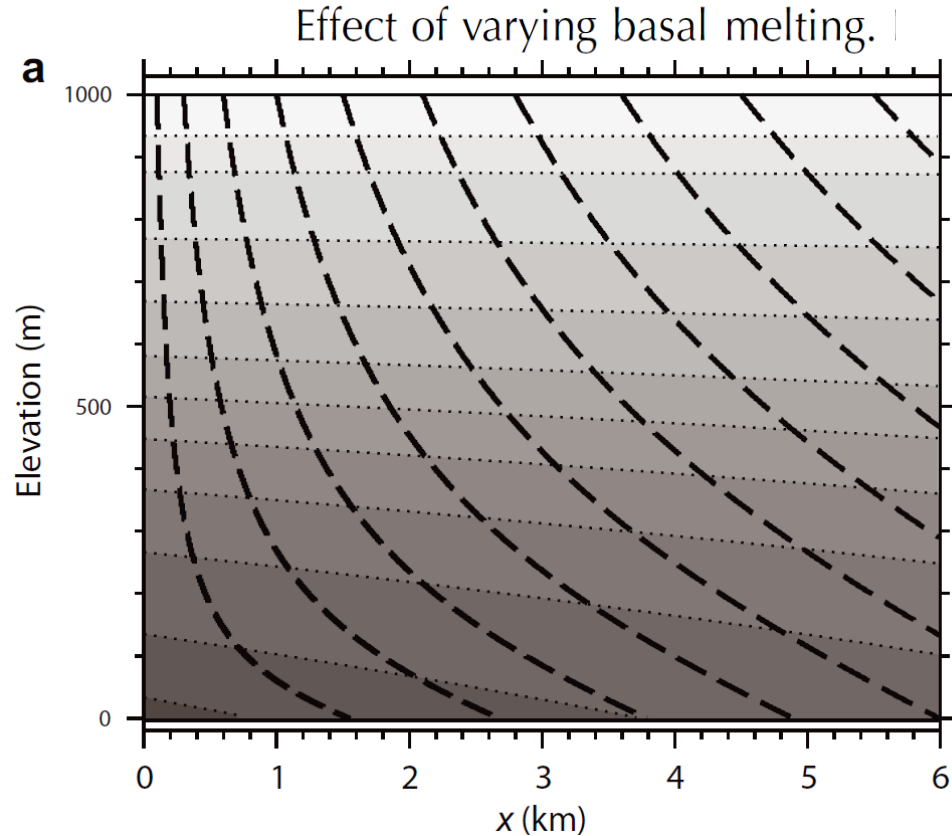
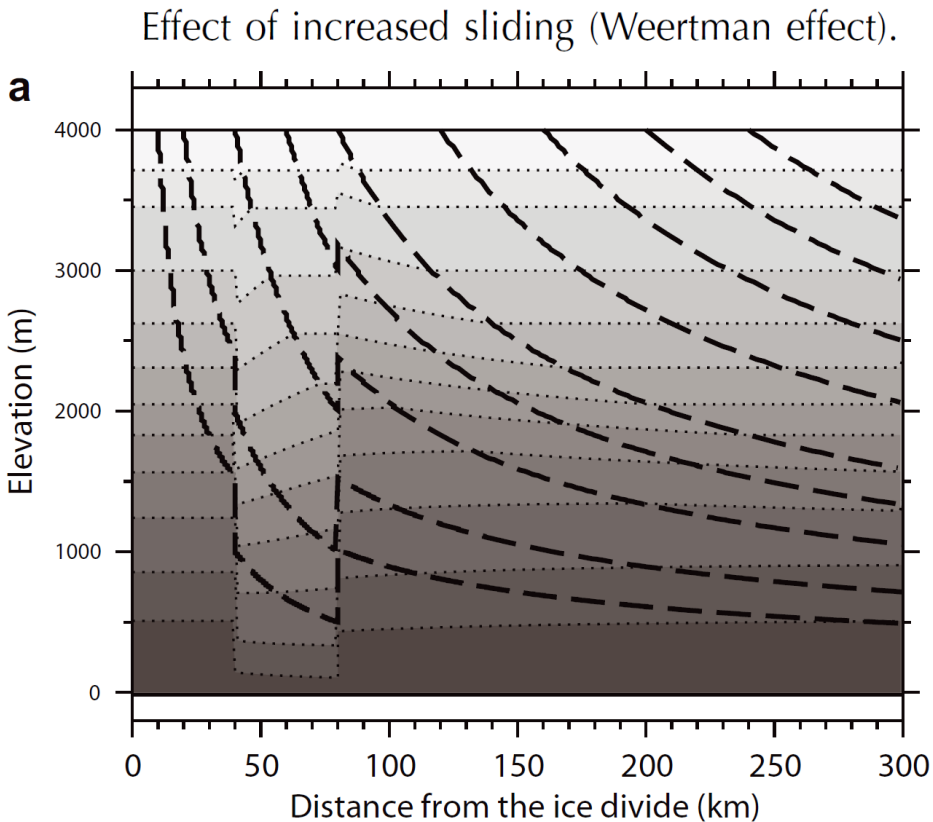
    # initial condition on age
    Age = np.zeros((time.size,depth.size))

    # main loop
    for timestep in np.arange(1,time.size):
        Age[timestep,1:] = Age[timestep-1,1:] + dt * \
            ((vertical_velocity[1:] + vertical_velocity[:-1])/2 * \
            (Age[timestep-1,1:] - Age[timestep-1,:-1])/grid_spacing + 1 )

    # return values
    Age_years = Age/seconds_per_year
    return Age_years, time, depth

# run the model
Age, time, depth = age(a_mperyear=0.2, p = 2)
```

Theoretical isochrones in 2D

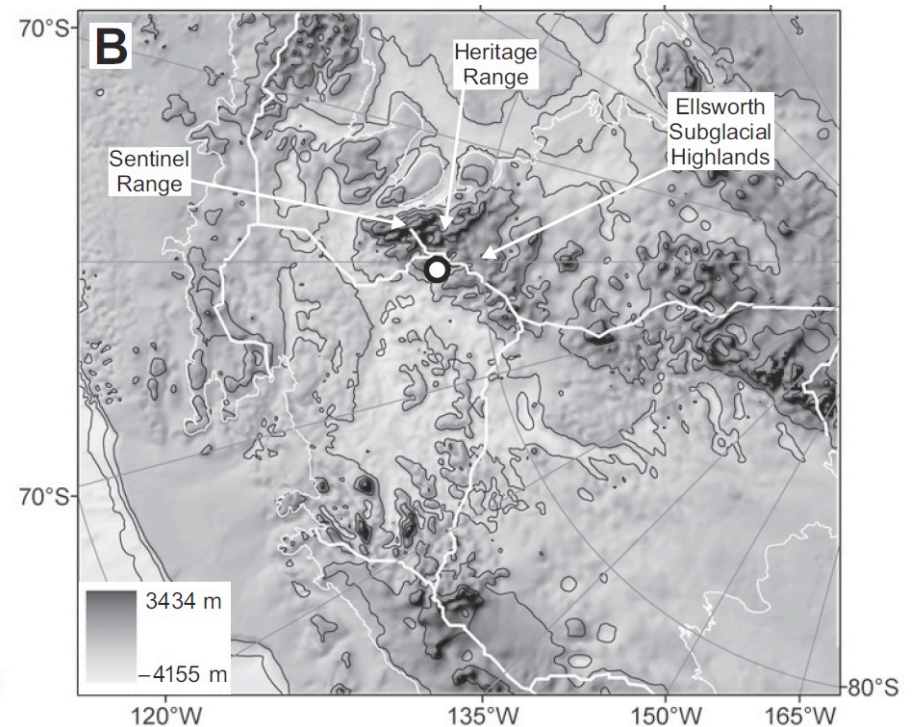
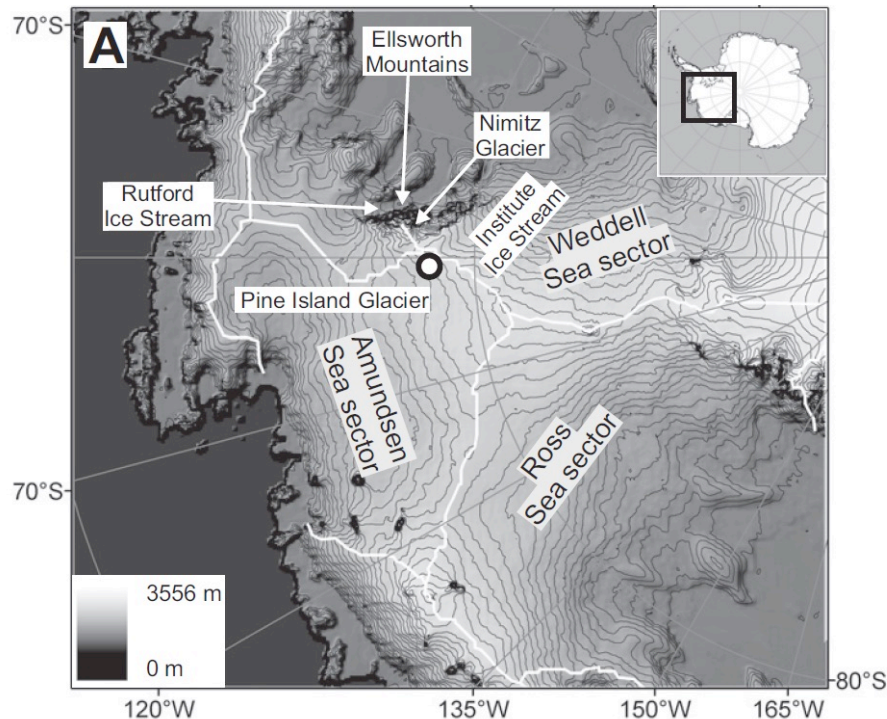


Parrenin and Hindmarsh (2007)

Qualitative interpretations of radar-observed isochrones

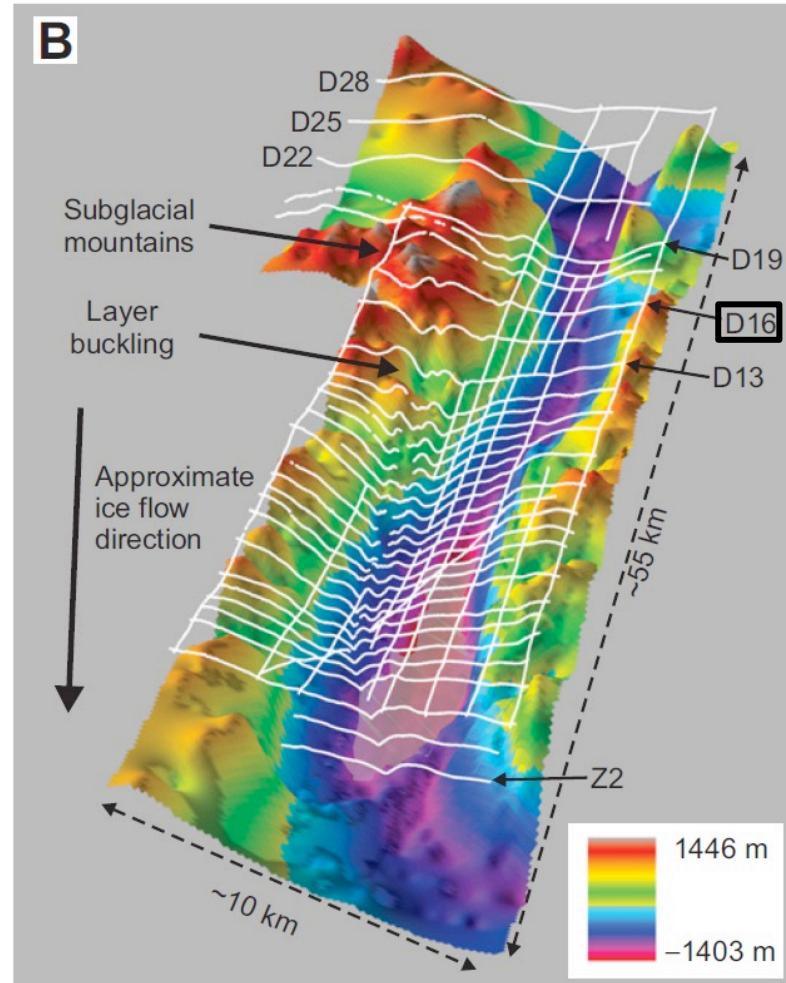
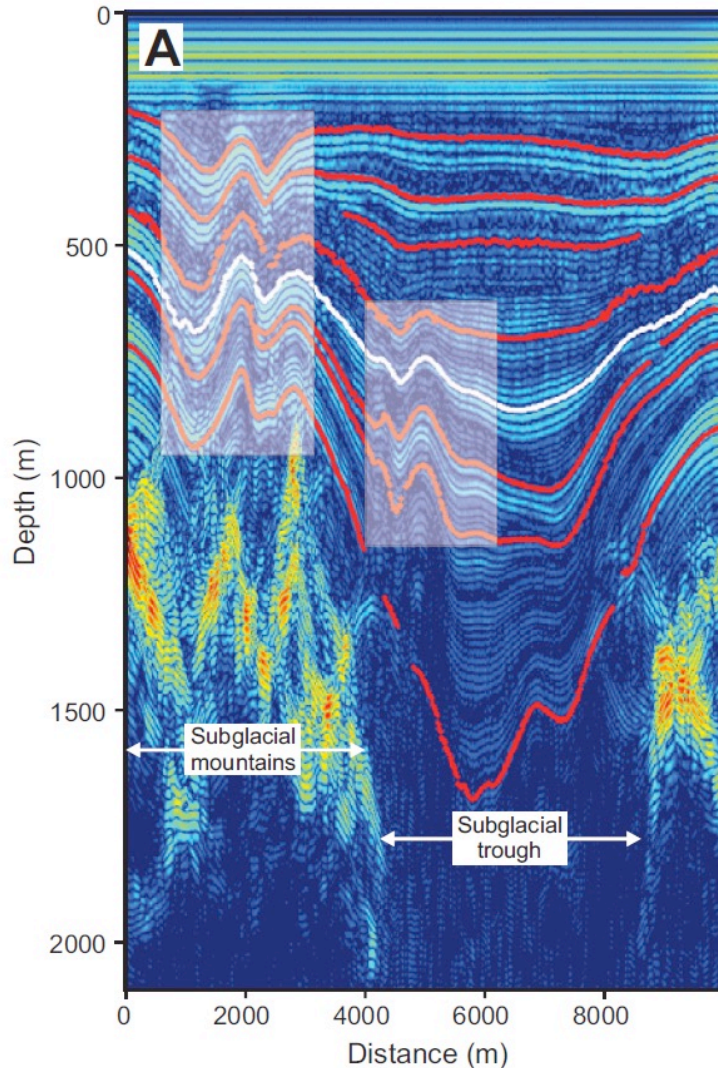
Holocene stability of the Amundsen-Weddell ice divide,
West Antarctica

N. Ross¹, M.J. Siegert¹, J. Woodward², A.M. Smith³, H.F.J. Corr³, M.J. Bentley⁴, R.C.A. Hindmarsh³, E.C. King³, and A. Rivera⁵



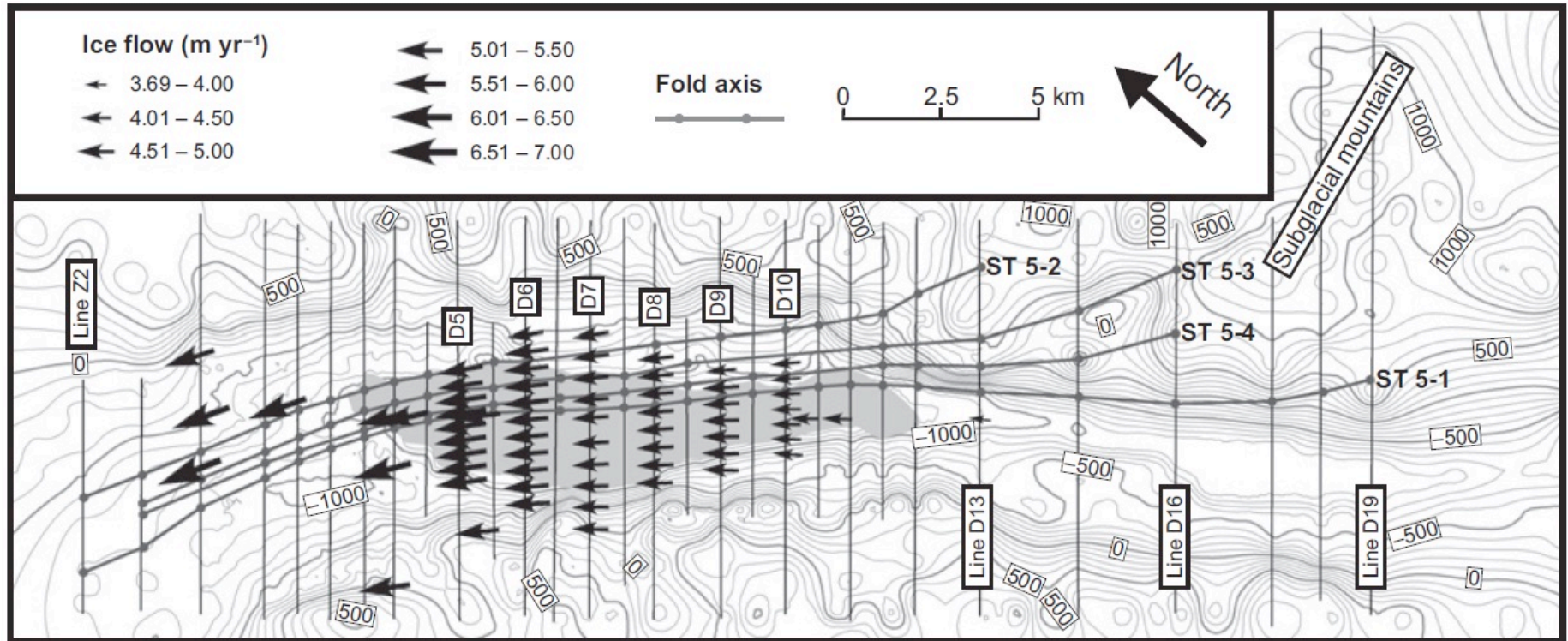
Ross et al. (2011)

Qualitative interpretations of radar-observed isochrones



Ross et al. (2011)

Qualitative interpretations of radar-observed isochrones



“ The internal layering around Ellsworth Subglacial Lake indicates that the configuration of ice flow in this sector of the interior WAIS has been stable for ~ 7000 yr. ”

Ross et al. (2010)

Quantitative analysis of radar-observed isochrones

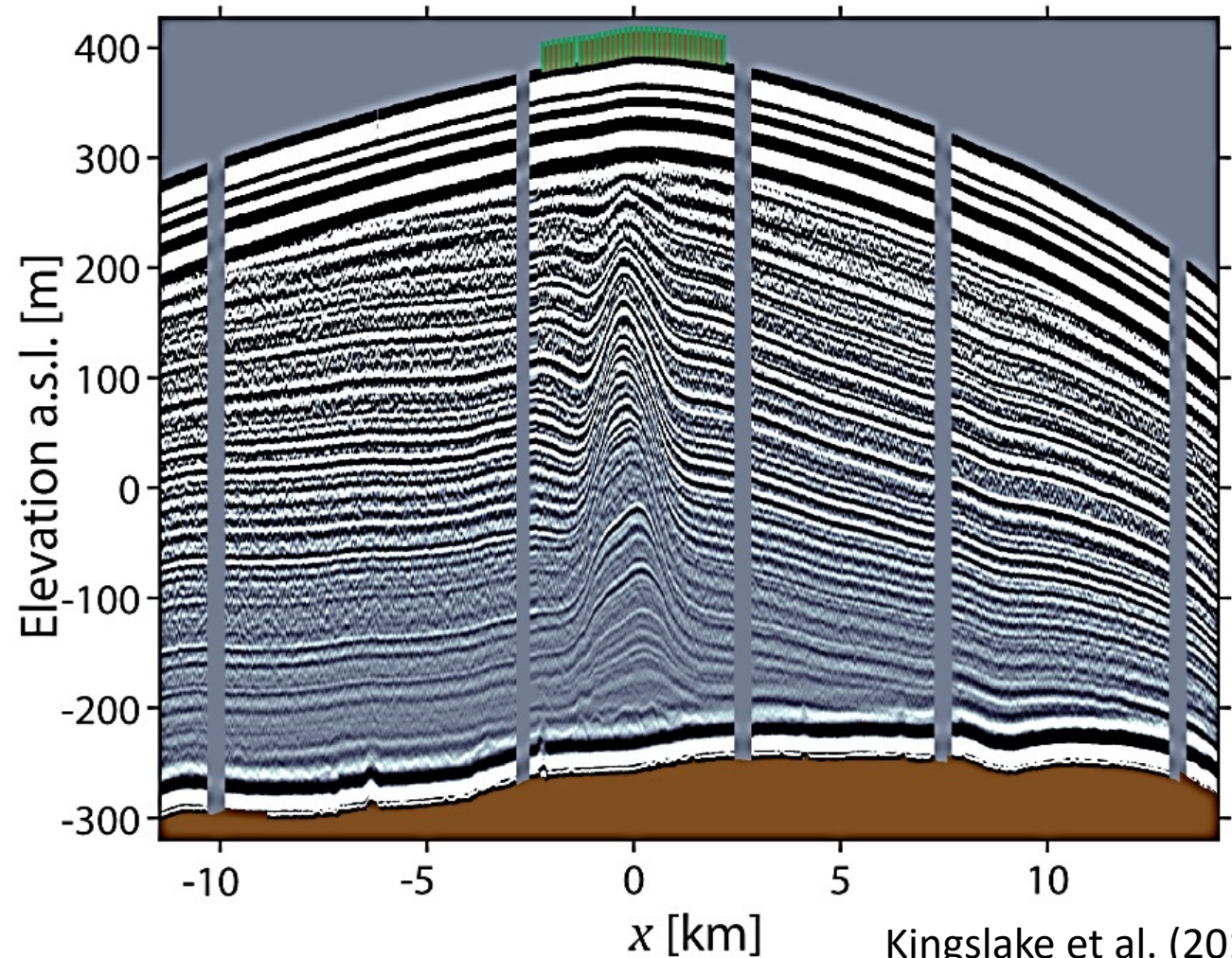
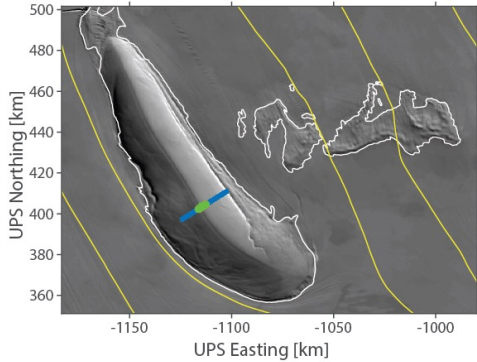
Ice-flow reorganization in West Antarctica 2.5 kyr ago dated using radar-derived englacial flow velocities

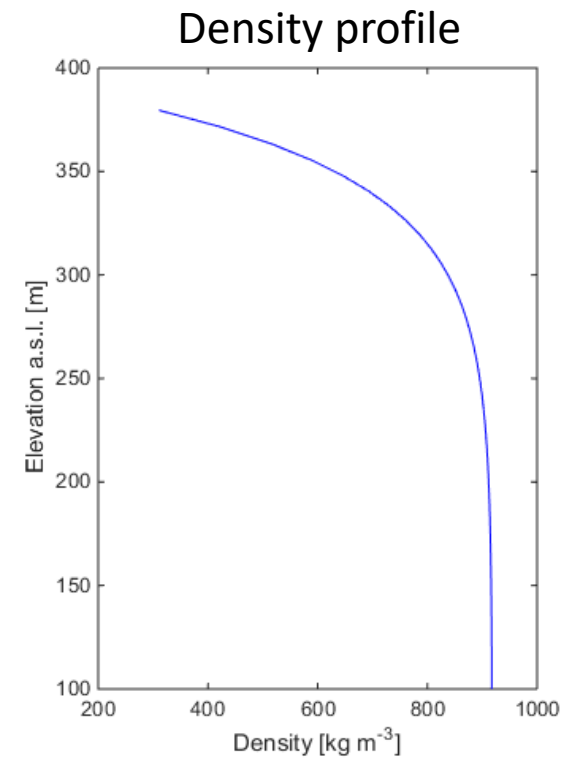
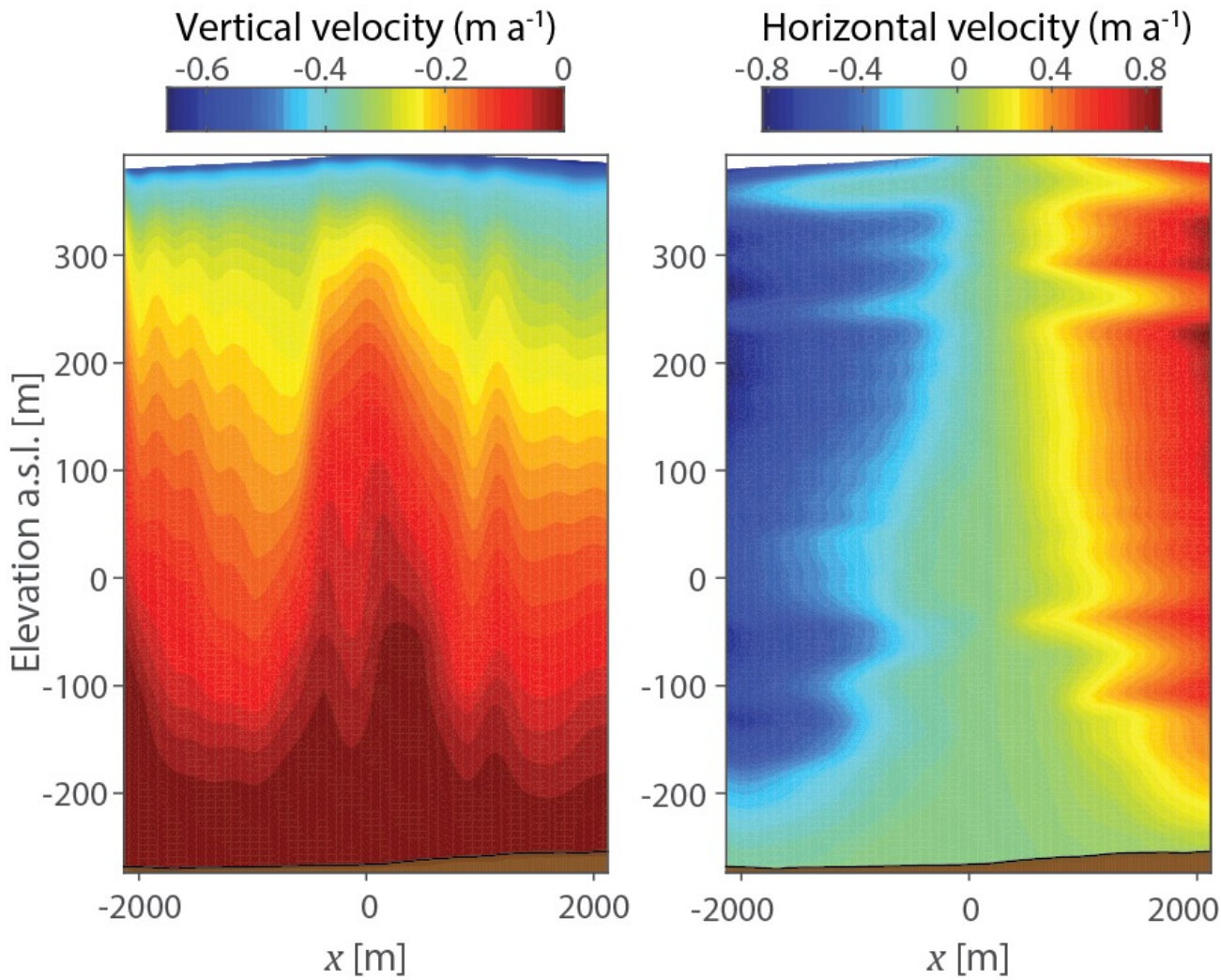
Jonathan Kingslake^{1,2}, Carlos Martín¹, Robert J. Arthern¹, Hugh F. J. Corr¹, and Edward C. King¹

Kingslake et al. (2016)

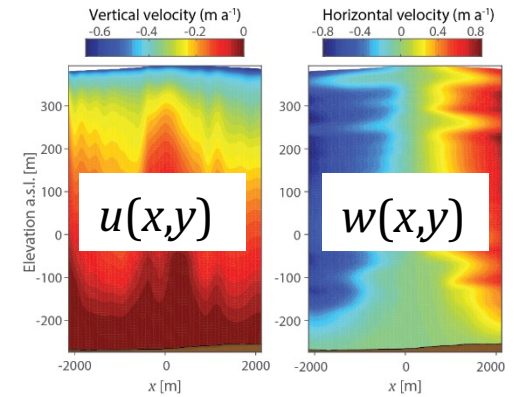
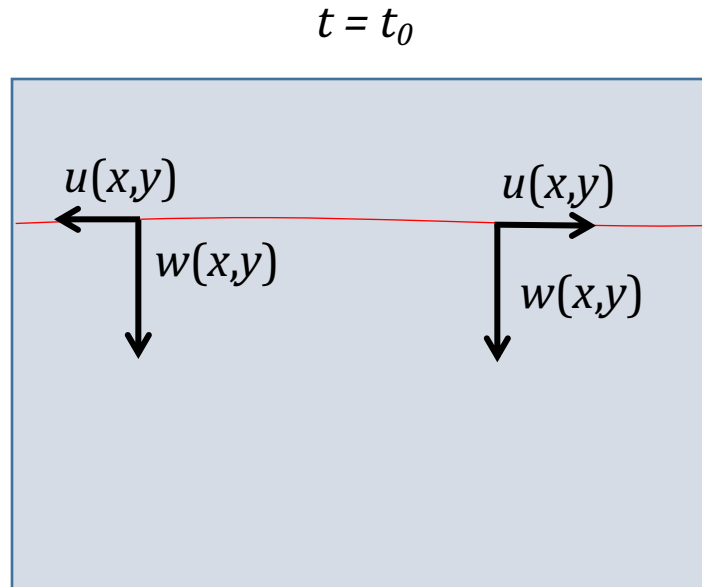
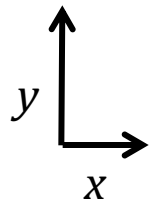


Quantitative analysis of radar-observed isochrones

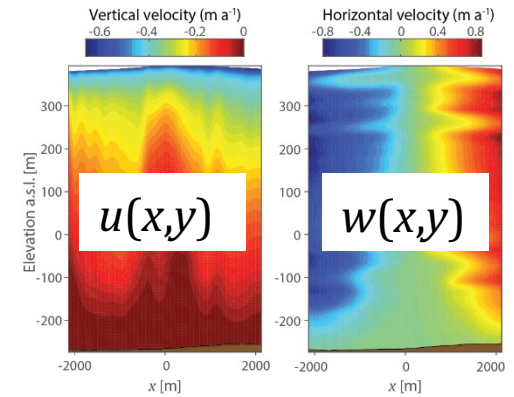
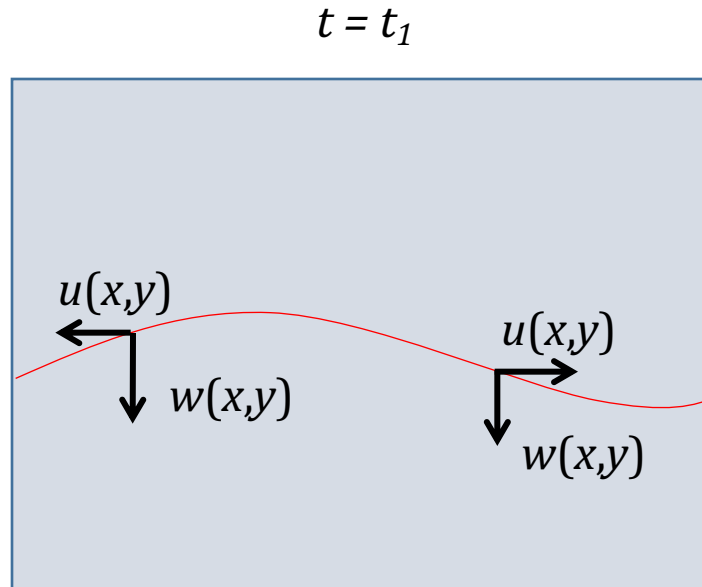
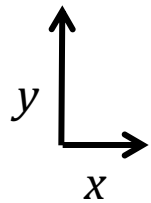




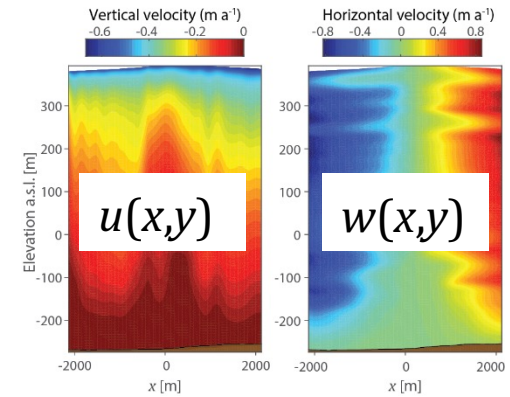
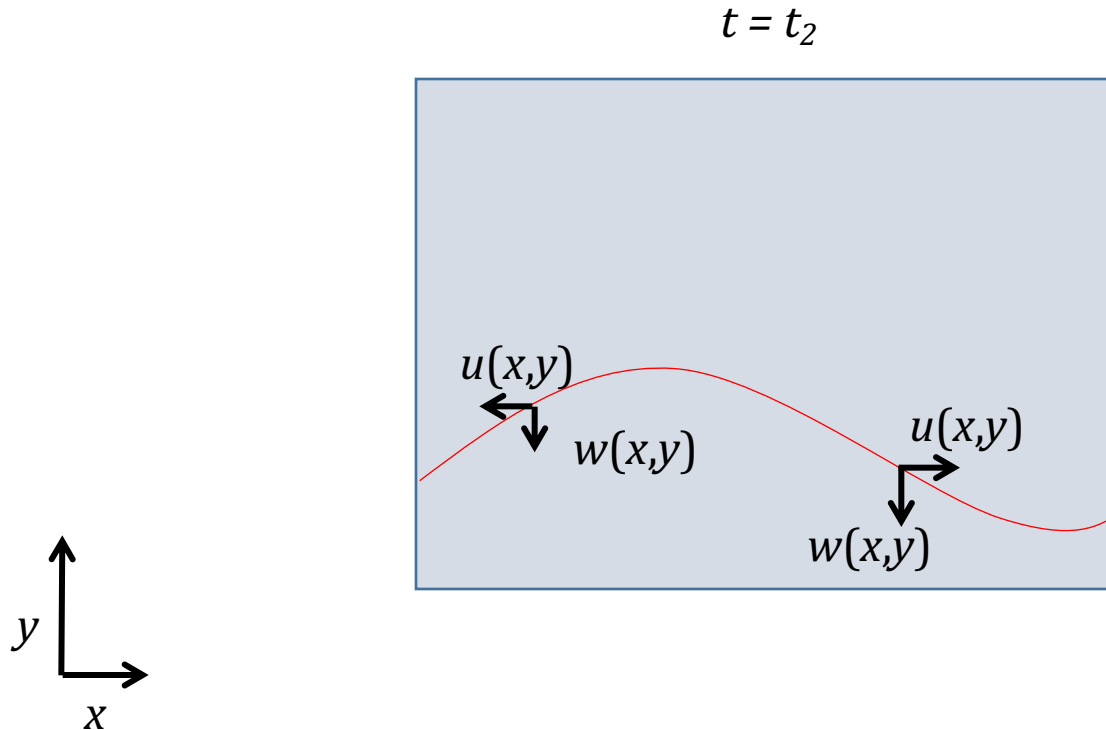
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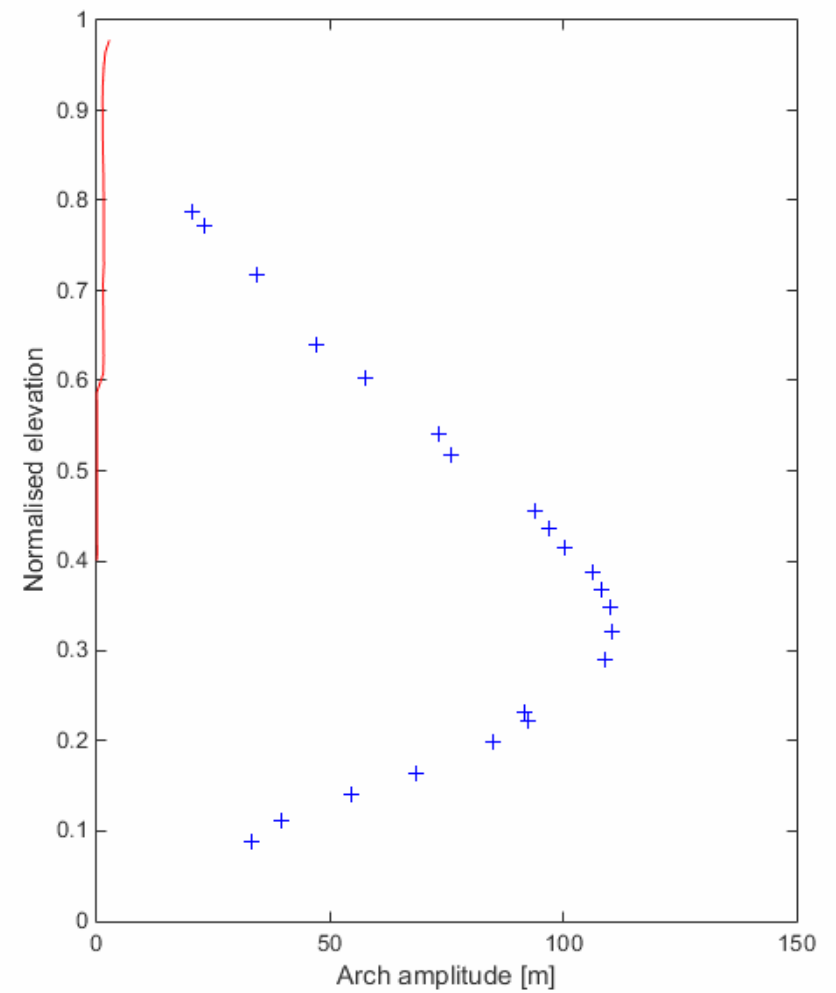
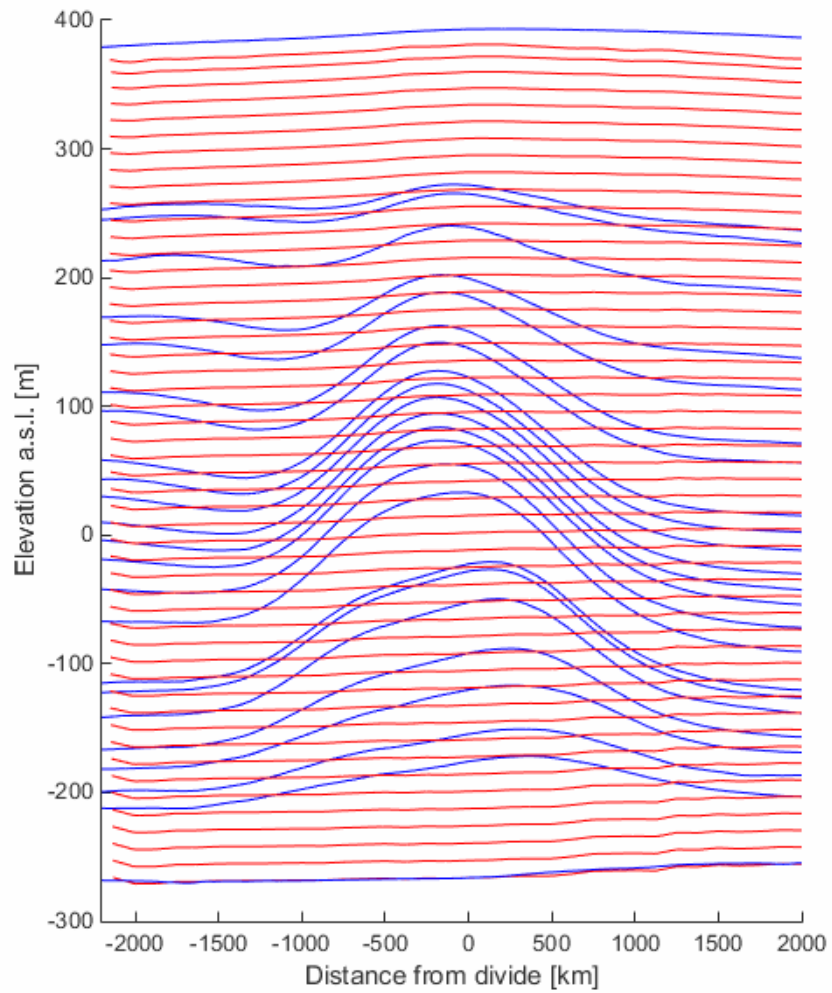
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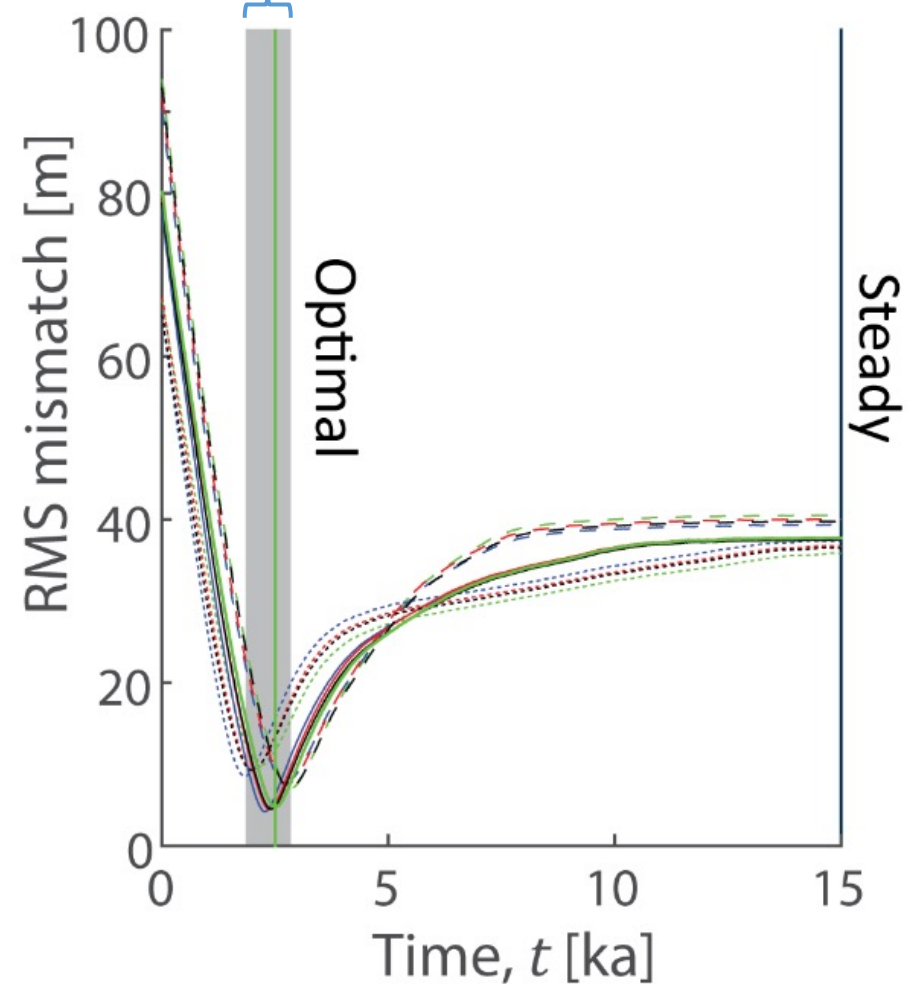
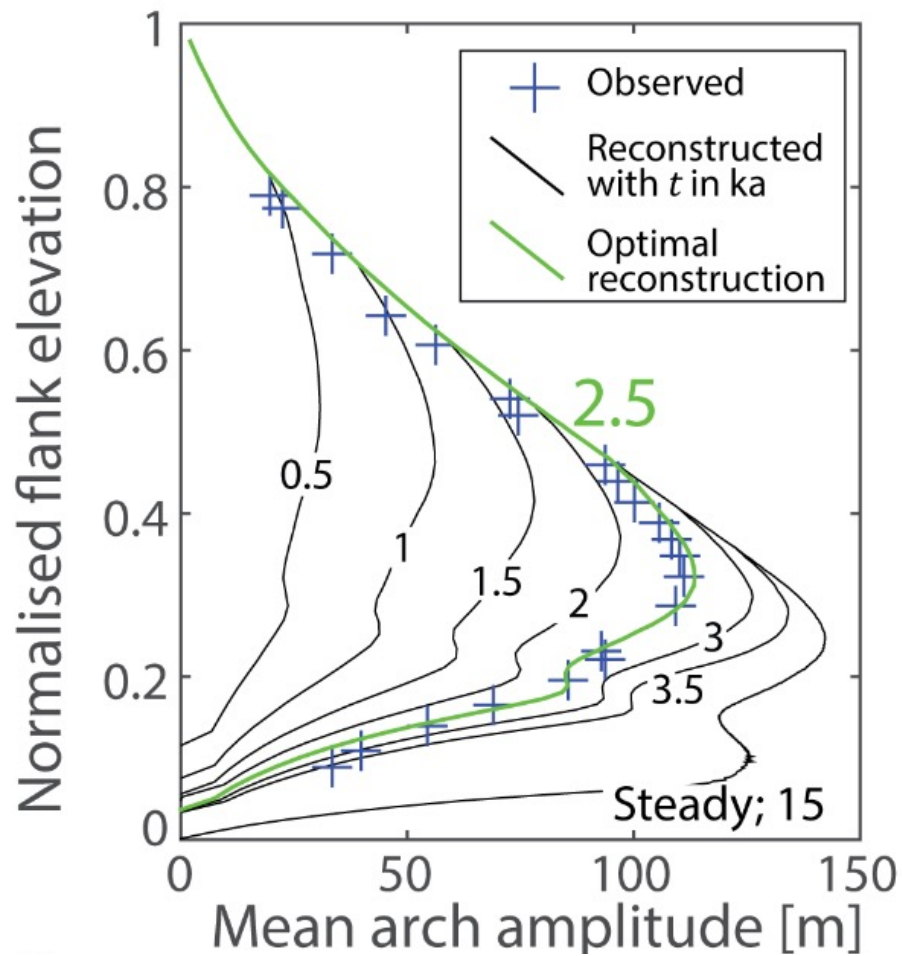
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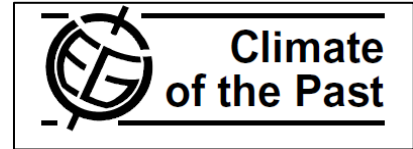
Reconstruction time = 0 a



1.8-2.9 ka BP



Ice core dating



The EDC3 chronology for the EPICA Dome C ice core

F. Parrenin¹, J.-M. Barnola¹, J. Beer², T. Blunier³, E. Castellano⁴, J. Chappellaz¹, G. Dreyfus⁵, H. Fischer⁶, S. Fujita⁷, J. Jouzel⁵, K. Kawamura⁸, B. Lemieux-Dudon¹, L. Loulergue¹, V. Masson-Delmotte⁵, B. Narcisi⁹, J.-R. Petit¹, G. Raisbeck¹⁰, D. Raynaud¹, U. Ruth⁶, J. Schwander³, M. Severi⁴, R. Spahni³, J. P. Steffensen¹¹, A. Svensson¹¹, R. Udisti⁴, C. Waelbroeck¹, and E. Wolff¹²

age marker	depth (m)	age (kyr BP)	error bar (kyr BP)
Krakatua	8.35	0.066	0.001
Tambora	12.34	0.134	0.001
Huaynaputina	23.20	0.349	0.001
Kuwae	29.27	0.492	0.005
El Chichon?	38.12	0.691	0.005
Unidentified	39.22	0.722	0.006
Unknown	41.52	0.779	0.006
¹⁰ Be/ ¹⁴ C	107.83	2.716	0.05
¹⁰ Be/ ¹⁴ C	181.12	5.28	0.05
YD/Holocene	361.5	11.65	0.18
PB/BO	427.2	15.0	0.24
¹⁰ Be peak	740.08	41.2	1
Mt Berlin erupt.	1265.10	92.5	2
term. II	1698.91	130.1	2
air content	1082.34	70.6	4
air content	1484.59	109.4	4
air content	1838.09	147.6	4
air content	2019.73	185.3	4
air content	2230.71	227.3	4
air content	2387.95	270.4	4
air content	2503.74	313.4	4
air content	2620.23	352.4	4
air content	2692.69	390.5	4
air content	2789.58	431.4	4
¹⁸ O _{atm}	2714.32	398.4	6

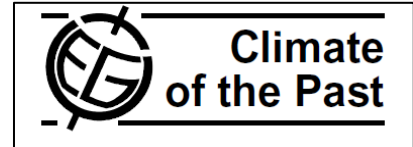
$$\text{age}(z) = \int_0^z \frac{1}{T(z') a(z')} dz'.$$

This contains all the
ice-flow physics

This contains the
climate history

Parrenin et al. (2007)

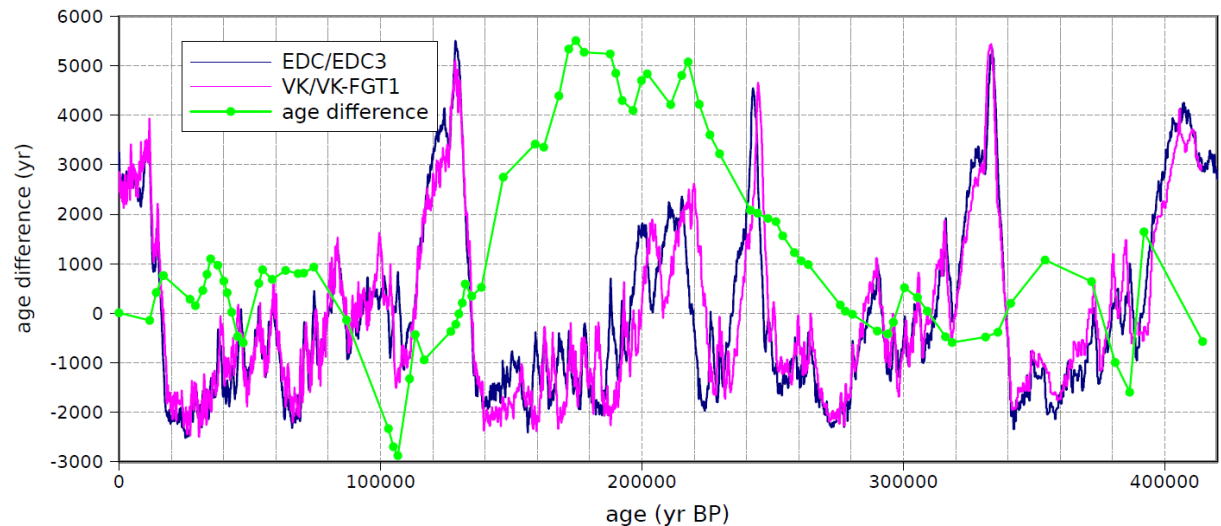
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Parrenin et al. (2007)

Summary

- Isochrones can be observed with radar
- Isochrones are age contours.
- Age can be treated as a scalar field, like temperature.
- Age structure is determined by present-day and past accumulation and ice flow.
- Many studies try to understand past accumulation and flow by observing present-day age structure.

References

- Kingslake, Jonathan, Carlos Martín, Robert J. Arthern, Hugh FJ Corr, and Edward C. King. "Ice-flow reorganization in West Antarctica 2.5 kyr ago dated using radar-derived englacial flow velocities." *Geophysical Research Letters* 43, no. 17 (2016): 9103-9112.
- Ross, N., Siegert, M.J., Woodward, J., Smith, A.M., Corr, H.F., Bentley, M.J., Hindmarsh, R.C., King, E.C. and Rivera, A., 2011. Holocene stability of the Amundsen-Weddell ice divide, West Antarctica. *Geology*, 39(10), pp.935-938.
- Parrenin, Frédéric, J-M. Barnola, J. Beer, Thomas Blunier, E. Castellano, J. Chappellaz, G. Dreyfus et al. "The EDC3 chronology for the EPICA Dome C ice core." *Climate of the Past* 3 (2007): 485-497.
- Parrenin, F. and Hindmarsh, R., 2007. Influence of a non-uniform velocity field on isochrone geometry along a steady flowline of an ice sheet. *Journal of Glaciology*, 53(183), pp.612-622.